

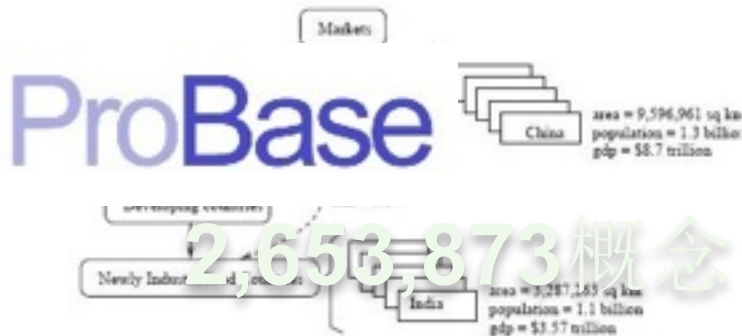
Natural Language Question Answering over Large-scale Linked Data

Kang Liu

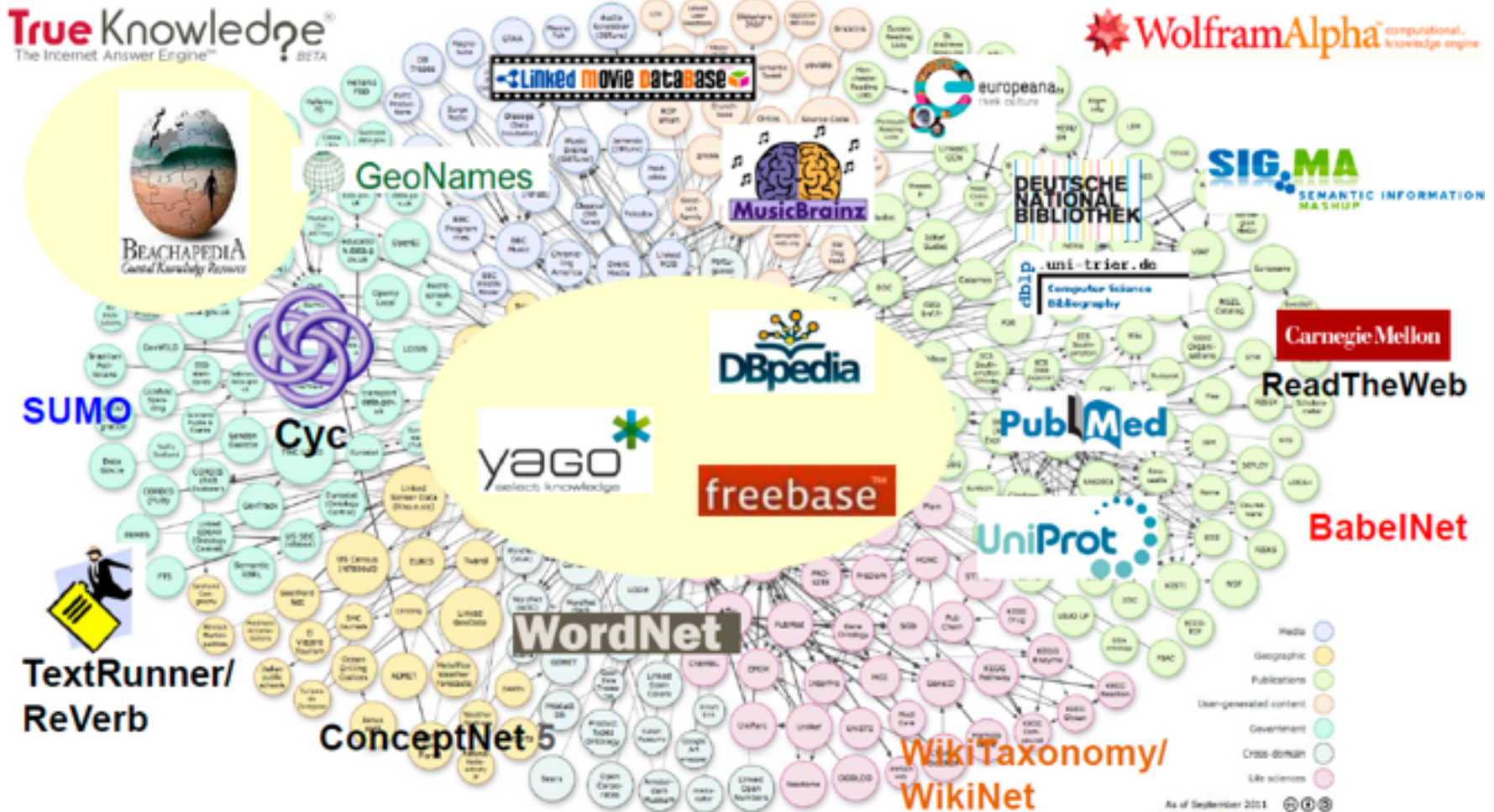
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8/30/2014 Kun Ming

Knowledge Graph: Linked Data



More Linked Data



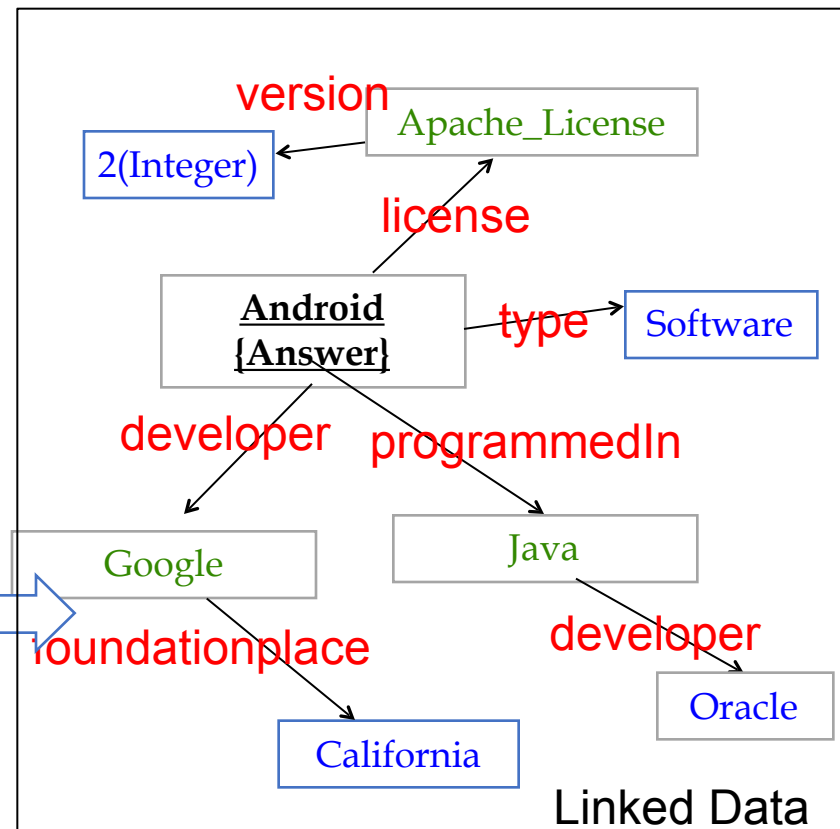
How to access these Linked Data

Which software has been developed by organizations founded in California, USA?

Experts or Developer

```
SELECT DISTINCT ?uri
WHERE {
  ?uri rdf:type dbo:Software.
  ?uri dbo:developer ?x1.
  ?x1 rdf:type dbo:Company.
  ?x1 dbo:foundationPlace dbr:California.
}
```

SPARQL



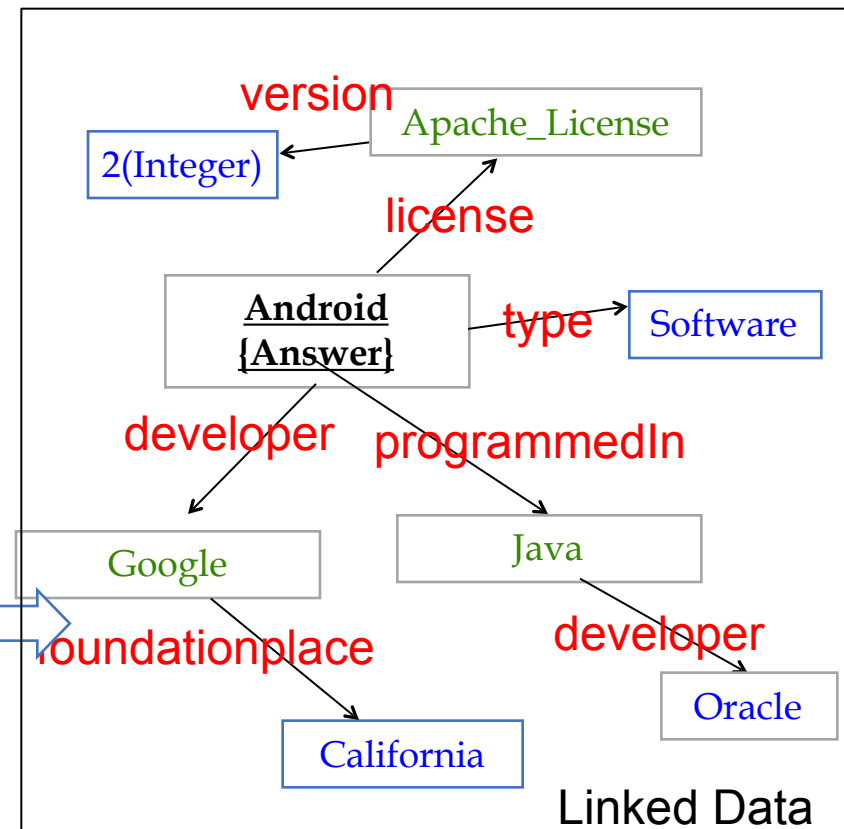
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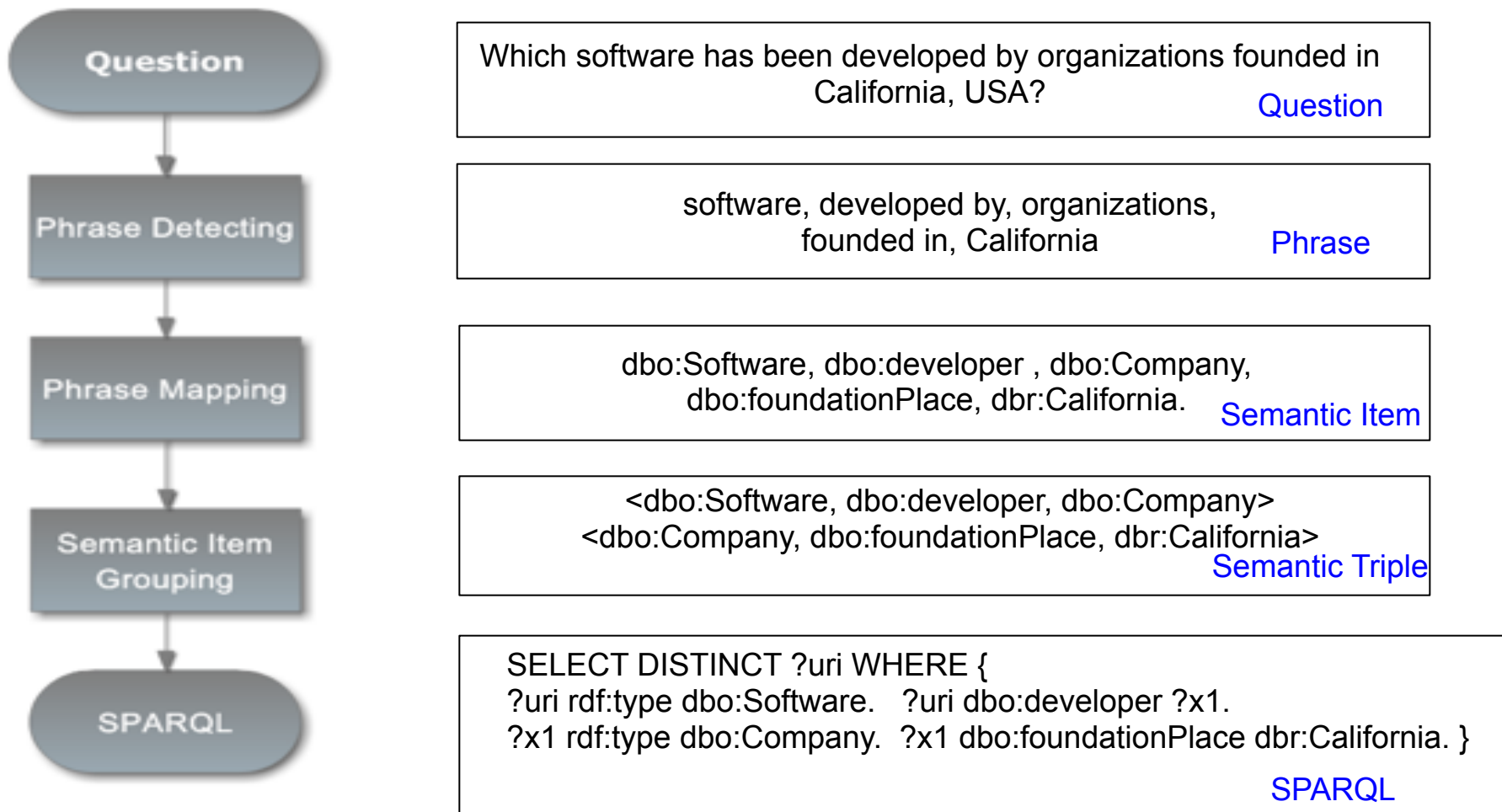
QA System

```
SELECT DISTINCT ?uri
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SPARQL

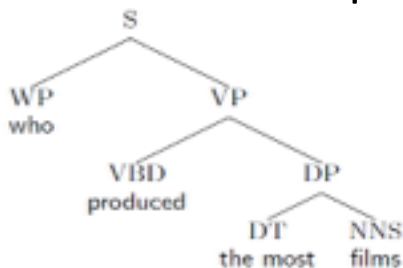


Pipeline Process



Challenges

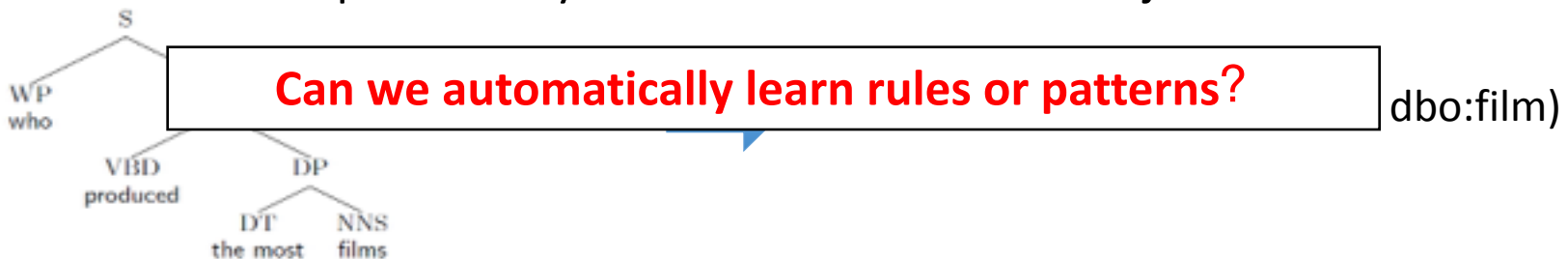
- Manually designed patterns
 - Phrase detection rules
 - NN|NNP: Entity
 - NN: Class | Property
 - VB: Property
 - Semantic item grouping patterns (syntactic patterns)
 - Verb and its arguments
 - Adjectives and its arguments
 - Prepositionally modified tokens and its objects



(?x, dbo:producer, dbo:film)

Challenges

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 - Phrase detection rules
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Challenges

Which software has been developed by organizations founded in California, USA?

- Ambiguities

- Phrase Detection:

- { California }, { California, USA }

- Phrase Mapping:

- California: {California_State}, {California_Film}

- Semantic Item Grouping:

- {dbo:Software, dbo:developer, dbo:Company}
 - {dbo:Software, dbo:foundationPlace, dbo:Company}

Challenges

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Can we make joint inference?

Our Solution

- Pattern Learning

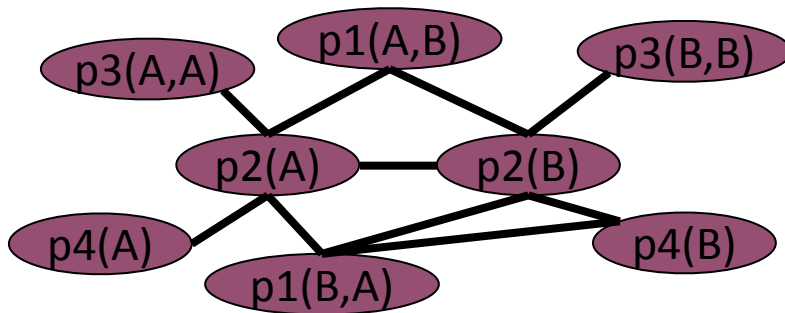
- Meta patterns

- Not only “verb and its arguments”
 - All syntactic paths maybe possible



- Joint Inference

- First-order logic formulas
 - Markov Logic Network



$$p(y) = \frac{1}{Z} \exp\left(\sum_{(\phi_i, w_i) \in L} w_i \sum_{c \in C^{\phi_i}} f_c^{\phi_i}(y) \right)$$

Predicates

- Hidden Predicates

<i>hasPhrase(i)</i>	The <i>i</i> th candidate phrase has been chosen
<i>hasResource(i, j)</i>	<i>The i</i> th phrase is mapped to the <i>j</i> th semantic item
<i>hasRelation (ri, rj, rr)</i>	The semantic item <i>ri</i> and <i>rj</i> can be grouped together with the relation type <i>rr</i>

- Observed Predicates

Describing the attributes of phrases and relation between two phrases	
<i>phraseIndex(p, i, j)</i>	The start and end position of phrase <i>p</i> in question.
<i>phrasePosTag(p, pt)</i>	The POS tag of head word in phrase <i>p</i> .
<i>phraseDepTag(p, q, dt)</i>	The dependency path tags between phrase <i>p</i> and <i>q</i> .
<i>phraseDepOne (p, q)</i>	If there is only one tag in the dependency path, the predicate is true.
<i>hasMeanWord (p, q)</i>	If there is any one meaning word in the dependency path of two phrases, the predicate is true.
Describing the attributes of semantic item and the mapping between phrase and semantic item	
<i>resourceType(r, rt)</i>	The type of semantic item <i>r</i> . Types of semantic items include <i>Entity</i> , <i>Class</i> and <i>Property</i>
<i>priorMatchScore(p, r, s)</i>	The prior score of phrase <i>p</i> mapping to semantic item <i>r</i> .
Describing the attributes of relation between two semantic items in knowledge base	
<i>hasRelatedness(p, q, s)</i>	The semantic coherence of semantic items.
<i>isTypeCompatible(p, q, rr)</i>	If semantic items <i>p</i> is type-compatible with semantic items <i>q</i> , the predicate is true.
<i>hasQueryResult(s, p, o, rr1, rr2)</i>	If the triple pattern consisting of semantic items <i>s, p, o</i> and relation types <i>rr1, rr2</i> have query results, the predicate is true.

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Formulas

- Hard Formulas

hf1	$hasPhrase(p) \Rightarrow hasResource(p, -)$
hf2	$hasResource(p, -) \Rightarrow hasPhrase(p)$
hf3	$ hasResource(p, -) \leq 1$
hf4	$!hasPhrase(p) \Rightarrow !hasResource(p, r)$
hf5	$hasResource(-, r) \Rightarrow hasRelation(r, -, -) \vee hasRelation(-, r, -)$
hf6	$ hasRelation(r1, r2, -) \leq 1$
hf7	$hasRelation(r1, r2, -) \Rightarrow hasResource(-, r1) \wedge hasResource(-, r2)$
hf8	$phraseIndex(p1, s1, e1) \wedge phraseIndex(p2, s2, e2) \wedge overlap(s1, e1, s2, e2) \wedge hasPhrase(p1) \Rightarrow !hasPhrase(p2)$
hf9	$resourceType(r, "Entity") \Rightarrow !hasRelation(r, -, "2.1") \wedge !hasRelation(r, -, "2.2")$
hf10	$resourceType(r, "Entity") \Rightarrow !hasRelation(-, r, "2.1") \wedge !hasRelation(r, -, "2.2")$
hf11	$resourceType(r, "Class") \Rightarrow !hasRelation(r, -, "2.1") \wedge !hasRelation(r, -, "2.2")$
hf12	$resourceType(r, "Class") \Rightarrow !hasRelation(-, r, "2.1") \wedge !hasRelation(r, -, "2.2")$
hf13	$!isTypeCompatible(r1, r2, rr) \Rightarrow !hasRelation(r1, r2, rr)$

Formulas

- Soft Formulas

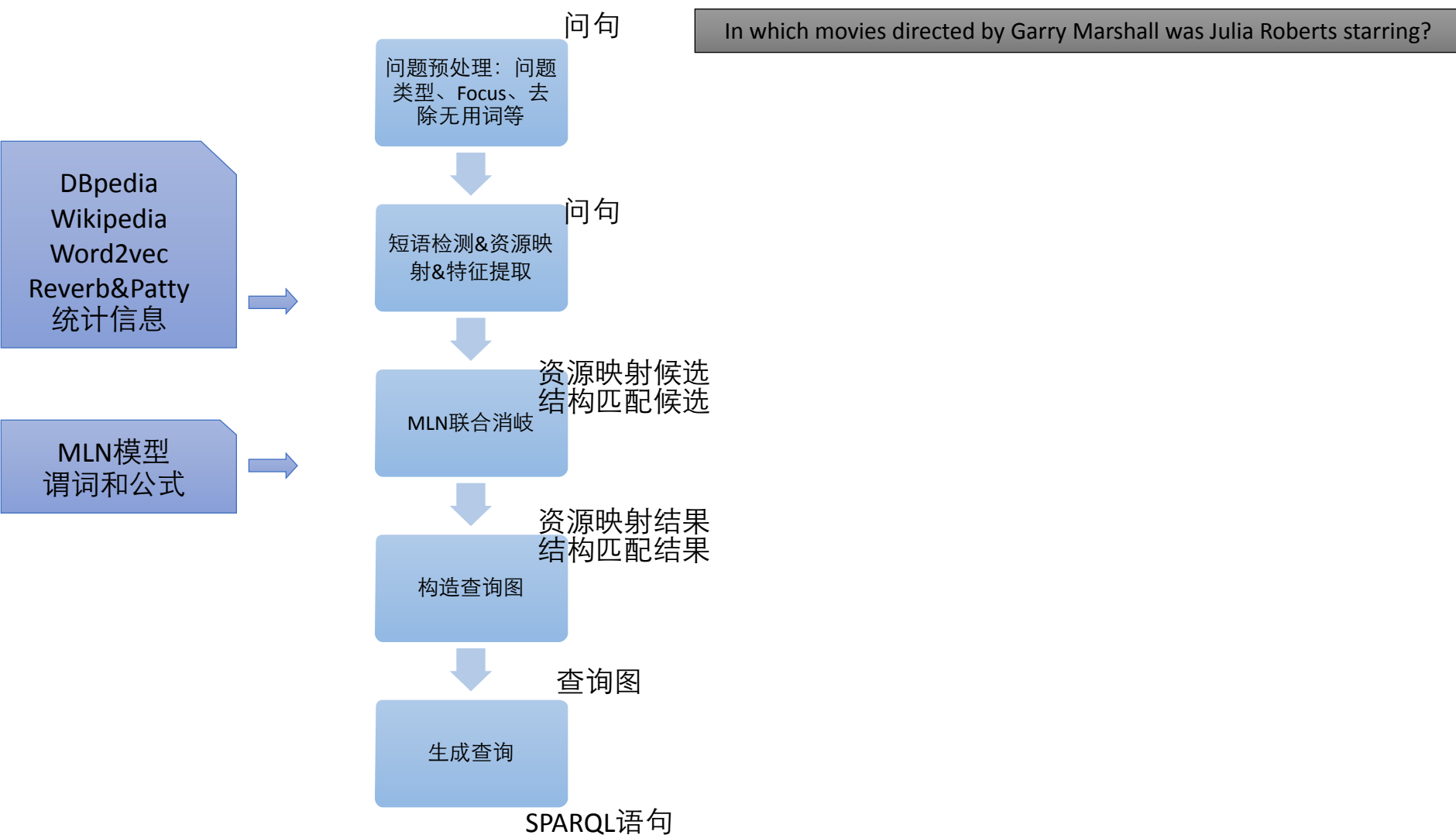
<i>sf1</i>	$\text{priorMatchScore}(p, r, s) \Rightarrow \text{hasPhrase}(p)$
<i>sf2</i>	$\text{priorMatchScore}(p, r, s) \Rightarrow \text{hasResource}(p)$
<i>sf3</i>	$\text{phrasePosTag}(p, pt+) \wedge \text{resourceType}(r, rt+) \Rightarrow \text{hasResource}(p, r)$
<i>sf4</i>	$\text{phraseDepTag}(p1, p2, dp+) \wedge \text{hasResource}(p1, r1) \wedge \text{hasResource}(p2, r2) \Rightarrow \text{hasRelation}(r1, r2, rr+)$
<i>sf5</i>	$\text{phraseDepTag}(p1, p2, dp+) \wedge \text{hasResource}(p1, r1) \wedge \text{hasResource}(p2, r2) \wedge \neg \text{hasMeanWord}(p1, p2) \Rightarrow \text{hasRelation}(r1, r2, rr+)$
<i>sf6</i>	$\text{phraseDepTag}(p1, p2, dp+) \wedge \text{hasResource}(p1, r1) \wedge \text{hasResource}(p2, r2) \wedge \text{phraseDepOne}(p1, p2) \Rightarrow \text{hasRelation}(r1, r2, rr+)$
<i>sf7</i>	$\text{hasRelatedness}(r1, r2, s) \wedge \text{hasResource}(_, r1) \wedge \text{hasResource}(_, r2) \Rightarrow \text{hasRelation}(r1, r2, _)$
<i>sf8</i>	$\text{hasQueryResult}(r1, r2, r3, rr1, rr2) \Rightarrow \text{hasRelation}(r1, r2, rr1) \wedge \text{hasRelation}(r2, r3, rr2)$

Formulas

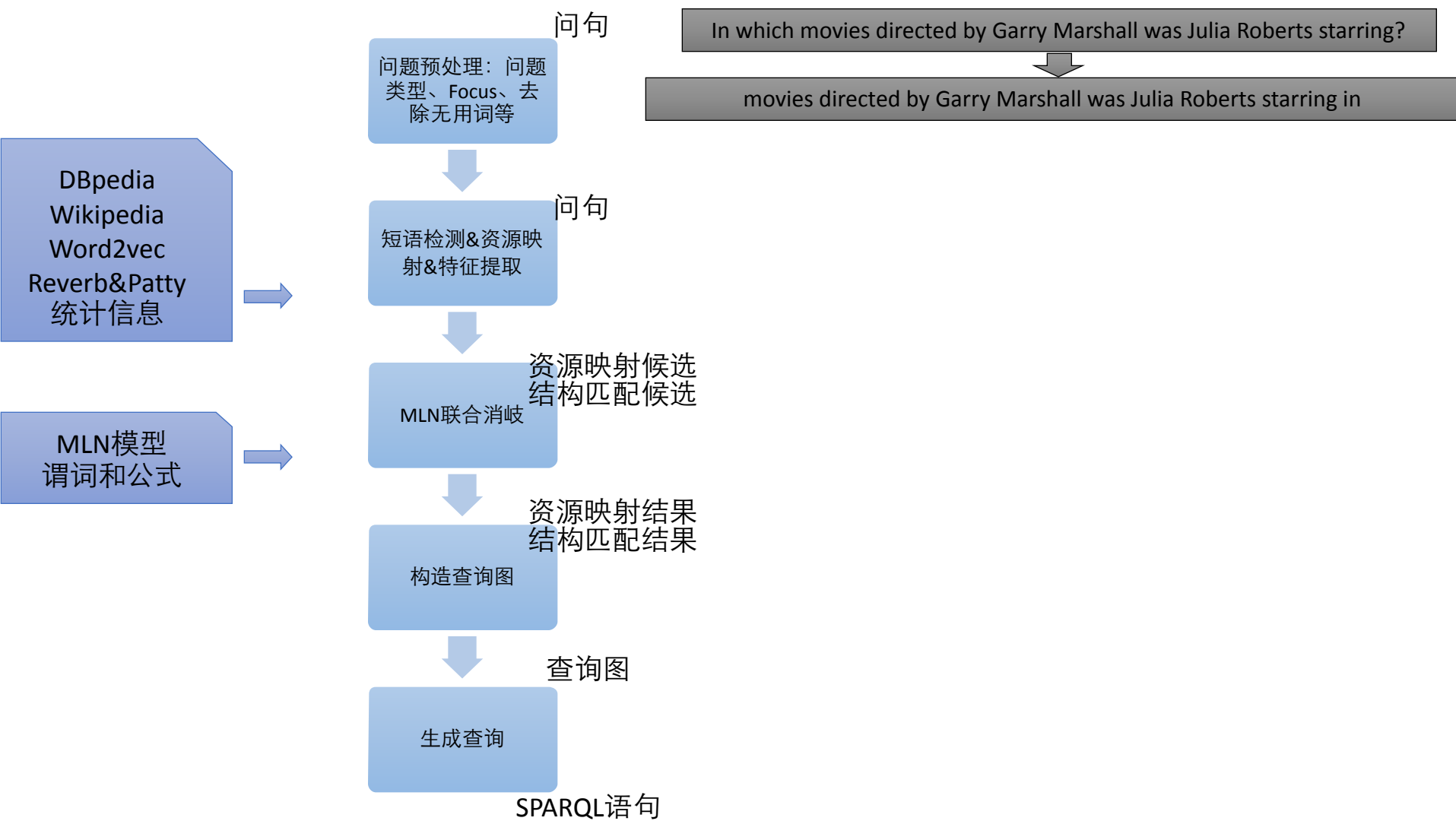
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<i>sf1</i>	$\text{priorMatchScore}(p, r, s) \Rightarrow \text{hasPhrase}(p)$
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<i>sf7</i>	$\text{hasRelatedness}(r1, r2, s) \wedge \text{hasResource}(-, r1) \wedge \text{hasResource}(-, r2) \Rightarrow \text{hasRelation}(r1, r2, -)$
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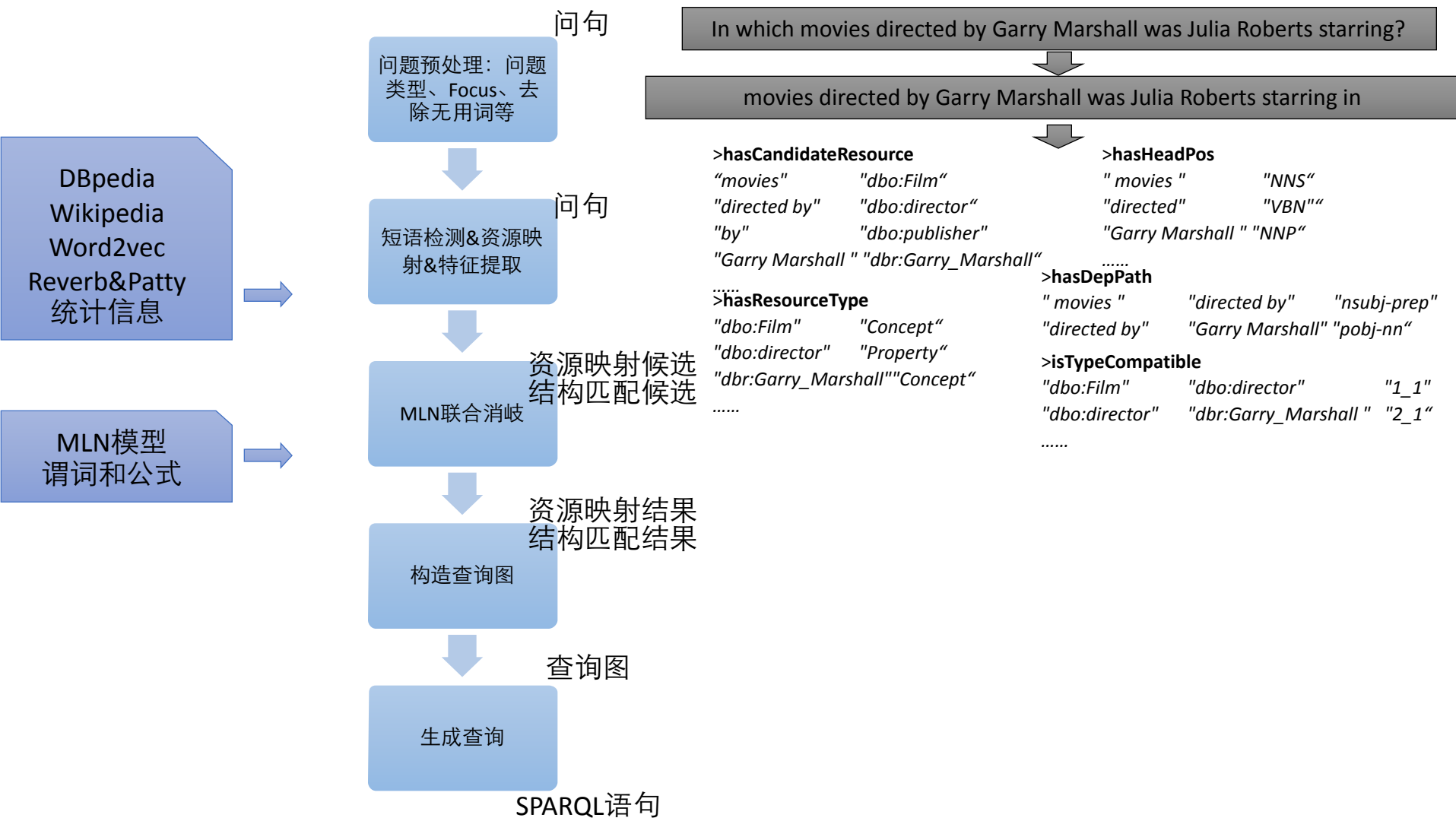
Framework



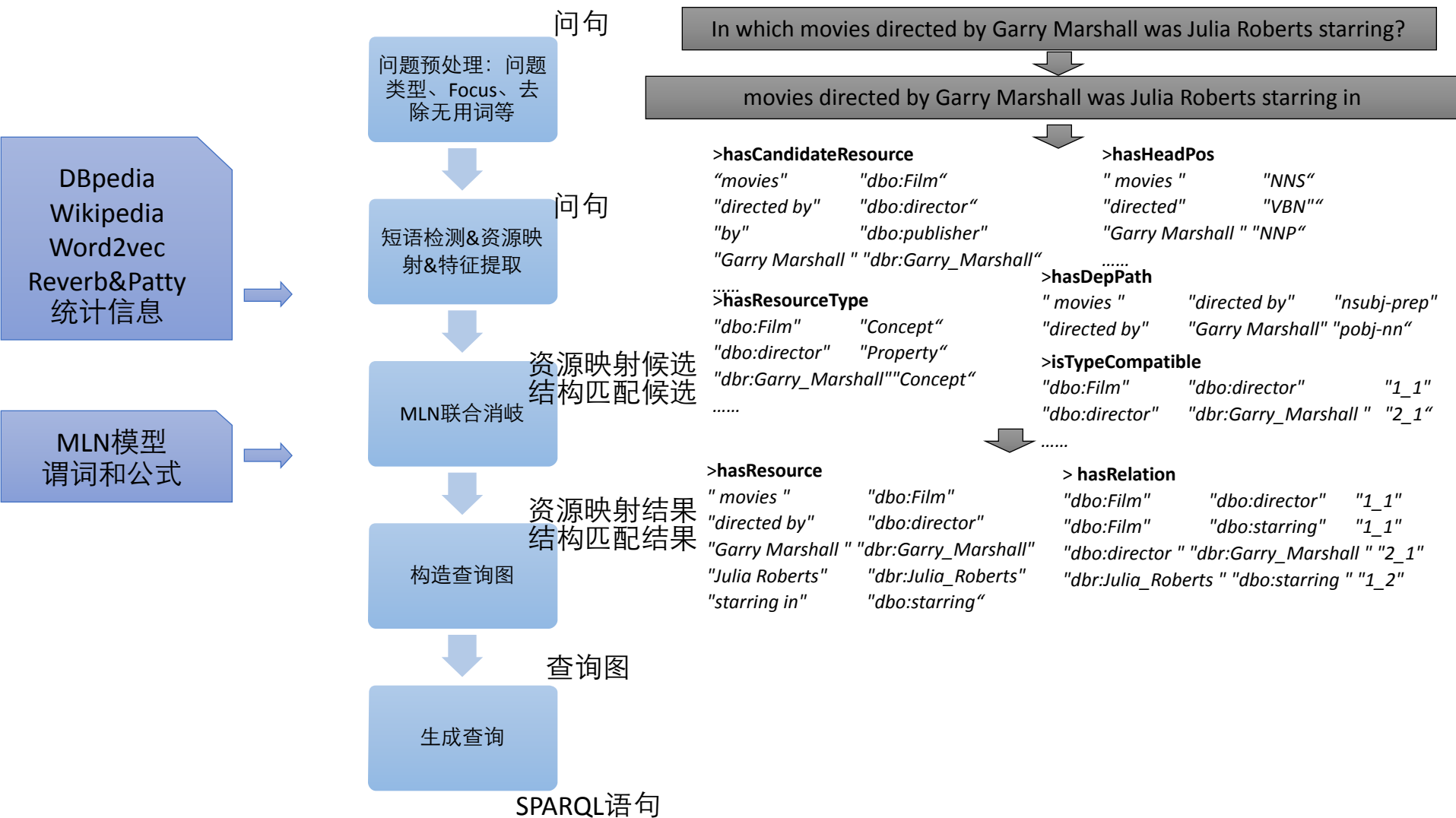
Framework



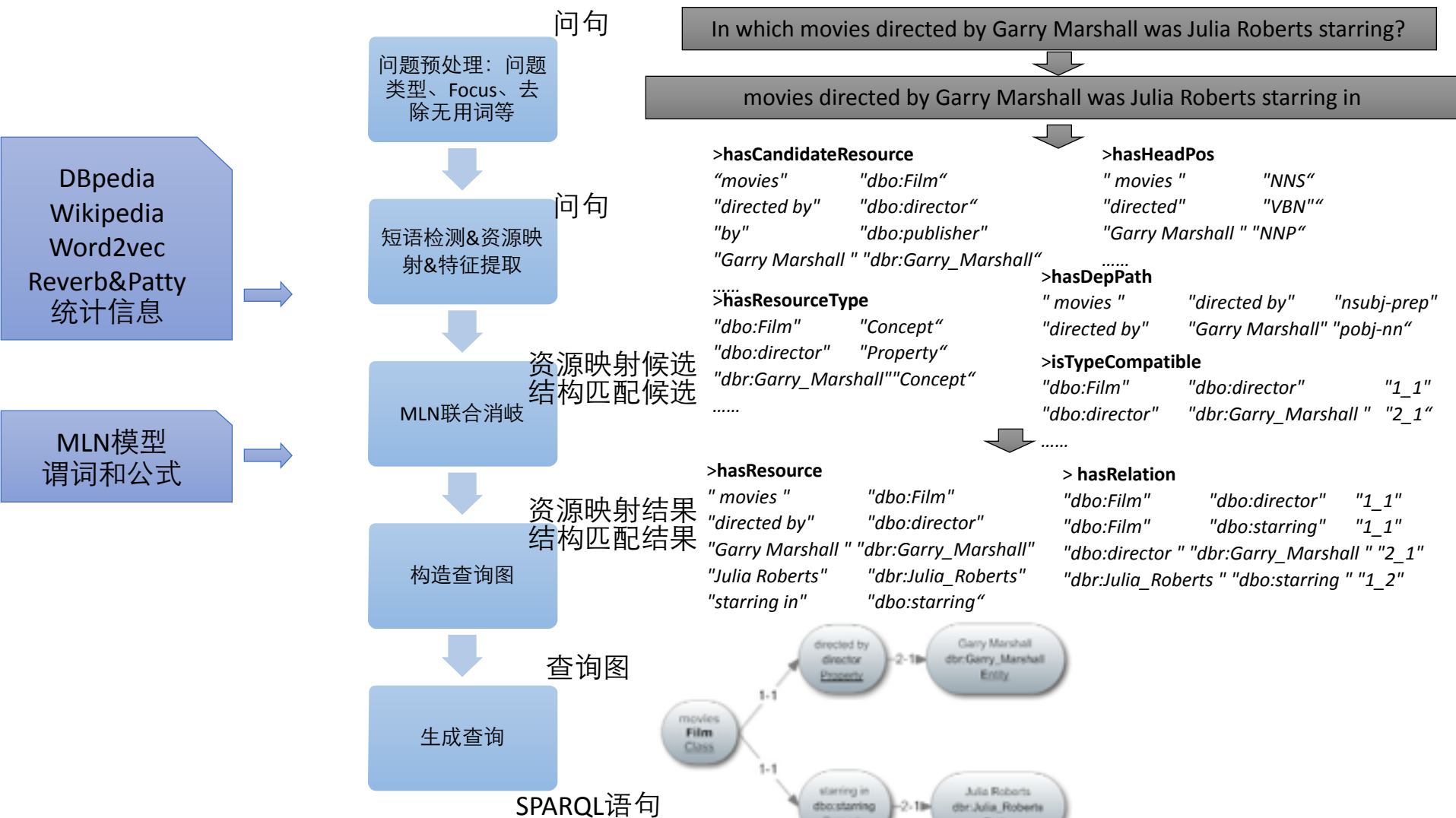
Framework



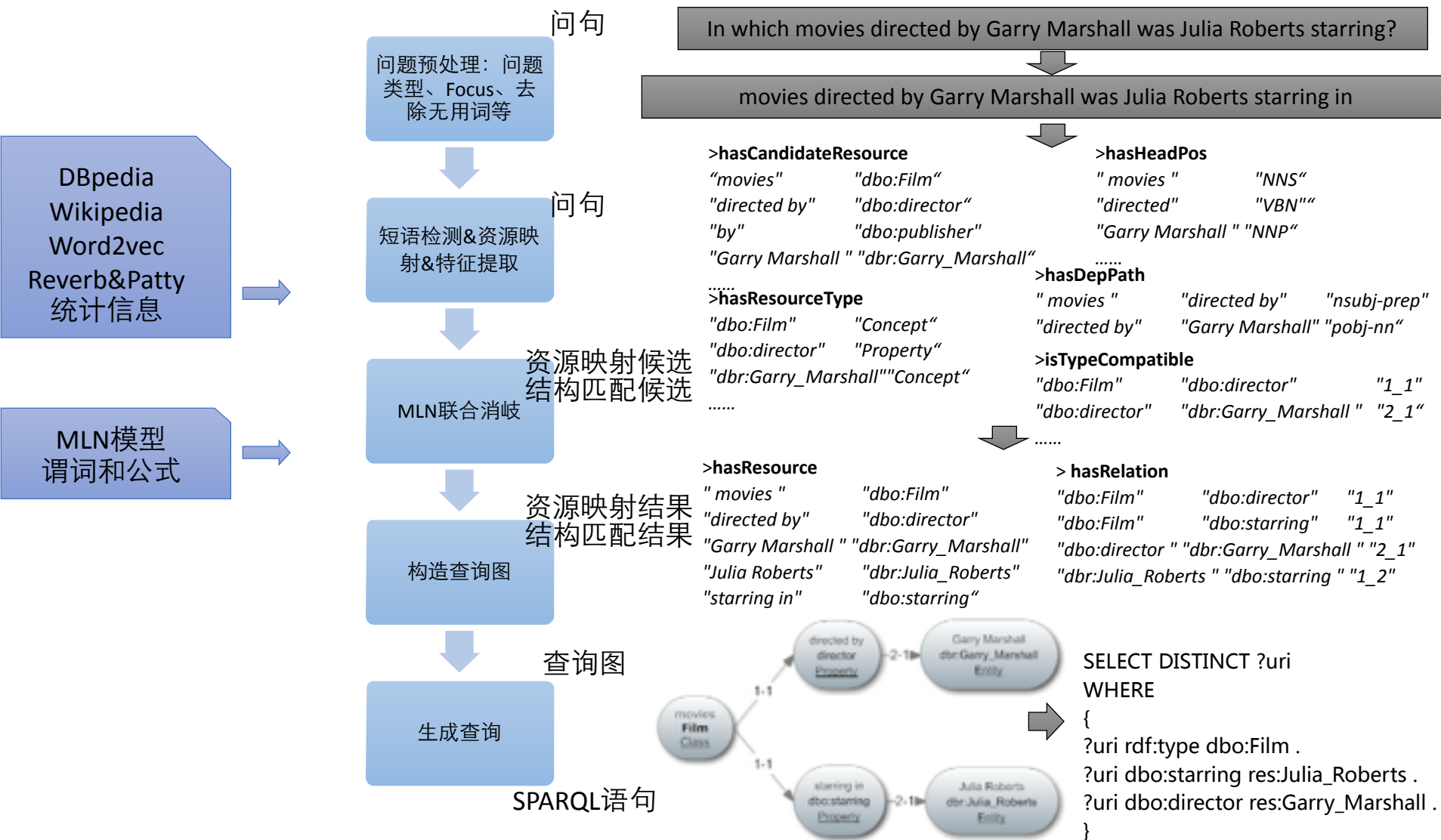
Framework



Framework

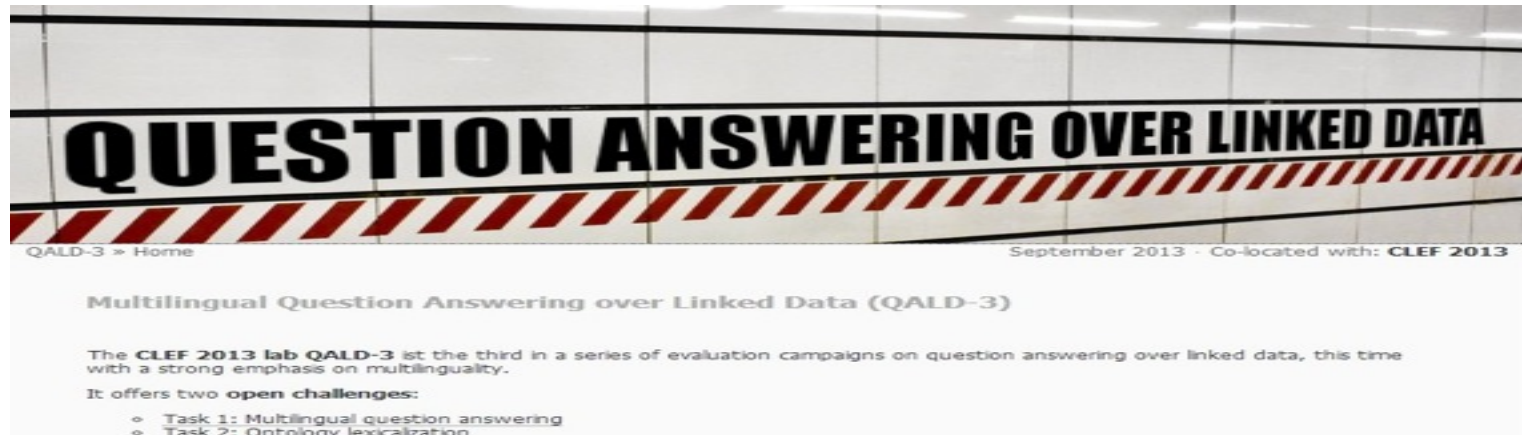


Framework



Experiments

- Questions
 - three collections of questions from QALD
 - QALD1, QALD3, QALD4
- Linked Data:
 - DBpedia, YAGO
- MLN: thebeast toolkit
 - inference algorithm: cutting plane approach[3]
 - weights learning algorithm: MIRA



Effect of Pattern Learning

<i>System</i>	#T	#Q	#A	P	R	F1
DEANNA (Yahya et al., 2012)	50	27	13	0.48	0.26	0.33
Ours	50	37	20	0.54	0.4	0.46

POS tag of Phrase	type of mapped Item	Weight
<i>nn</i>	<i>Entity</i>	2.11
<i>nn</i>	<i>Class</i>	0.243
<i>nn</i>	<i>Relation</i>	0.335
<i>vb</i>	<i>Relation</i>	0.517
<i>wp</i>	<i>Class</i>	0.143
<i>wr</i>	<i>Class</i>	0.025

Effect of Joint Inference

Benchmark	PD			PM			MG			QA					
	P	R	F1	P	R	F1	P	R	F1	#T	#Q	#A	P	R	F1
QALD-1(Joint)	0.93	0.981	0.955	0.895	0.944	0.919	0.703	0.813	0.754	50	37	20	0.54	0.4	0.46
QALD-1(Pipeline)	0.921	0.972	0.946	0.868	0.917	0.892	0.585	0.859	0.696	50	34	17	0.5	0.34	0.41
QALD-3(Joint)	0.941	0.941	0.941	0.878	0.918	0.898	0.636	0.798	0.708	99	75	45	0.6	0.46	0.52
QALD-3(Pipeline)	0.912	0.912	0.912	0.829	0.867	0.848	0.677	0.789	0.729	99	75	42	0.56	0.42	0.48
QALD-4(Joint)	0.947	0.978	0.963	0.937	0.967	0.952	0.776	0.865	0.817	50	26	15	0.58	0.3	0.4
QALD-4(Pipeline)	0.937	0.967	0.952	0.905	0.935	0.920	0.683	0.827	0.748	50	24	13	0.54	0.26	0.35

Effect of Joint Inference

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	P	R	F1	P	R	F1	P	R	F1	#T	#Q	#A	P	R	F1
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Ours vs. state-of-the-art

Test set	System	#T	#Q	#A	P	R	F1
QALD-3	CASIA (He et al., 2013)	99	52	29	0.56	0.3	0.38
	Scalewelis (Joris and Ferré, 2013)	99	70	32	0.46	0.32	0.38
	RTV (Cristina et al., 2013)	99	55	30	0.55	0.3	0.39
	Intui2 (Corina, 2013)	99	99	28	0.28	28	0.28
	SWIP (Pradel et al., 2013)	99	21	15	0.71	0.15	0.25
	Ours	99	75	45	0.6	0.46	0.52
QALD-4 ²⁰	gAnswer	50	25	16	0.64	0.32	0.43
	Intui3	50	33	10	0.30	0.2	0.24
	ISOFT	50	50	10	0.2	0.2	0.2
	RO EU	50	50	6	0.12	0.12	0.12
	Ours	50	26	15	0.58	0.3	0.4

Conclusion and Future work

- Pattern learning is needed for parsing a question over large-scale linked data
- Joint inference can be effective for improving the performance of natural language question answering
- Scaled up to multiple interlinked knowledge bases
- Labeled data is insufficient to build up a robust model
- More robust solutions to find the implicit properties in questions

Reference:

Questioning Answering over Linked Data Using Markov First-order Logic,

To appear in Proceedings of EMNLP 2014, Doha, Qatar, October, 25-29

Thanks

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